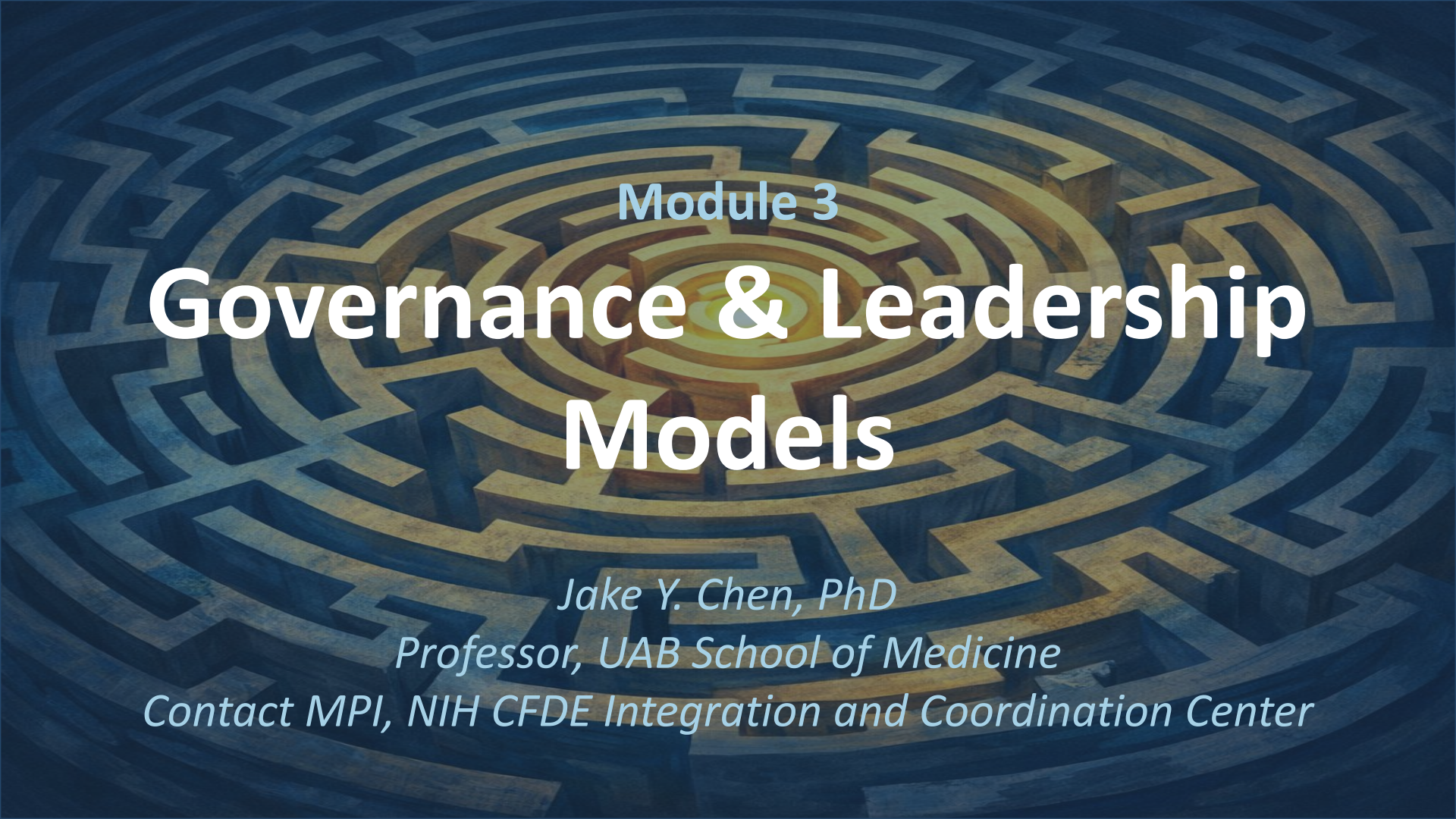




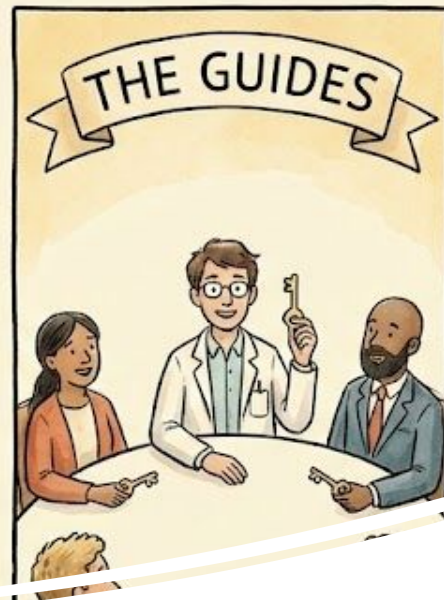
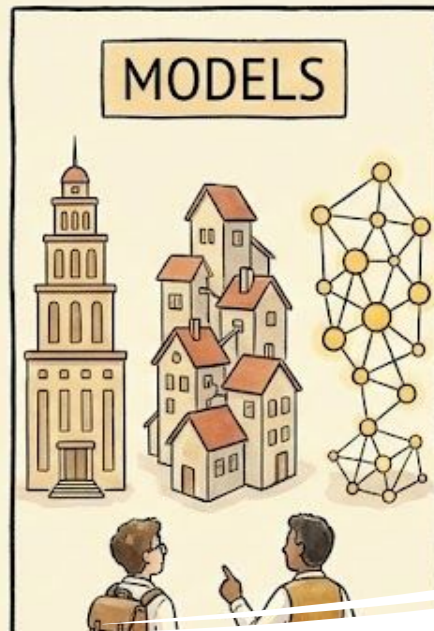
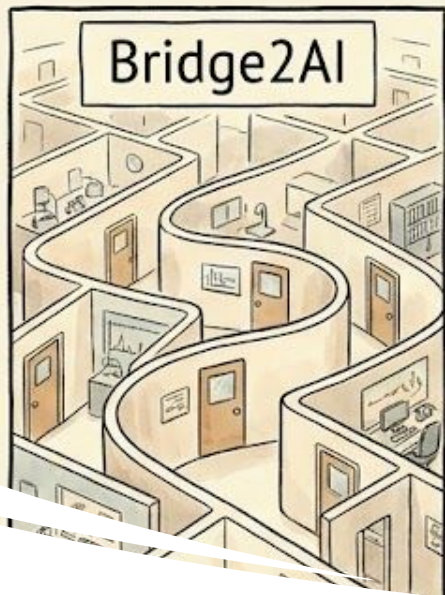
Module 3

Governance & Leadership Models

Jake Y. Chen, PhD

Professor, UAB School of Medicine

Contact MPI, NIH CFDE Integration and Coordination Center



Learning Objectives

- Understand the layered decision pathways in large consortia
- Compare governance models: hierarchy, distributed, and network
- Distinguish what is fixed vs. negotiable in governance structures
- Prepare to engage in a dynamic fishbowl discussion

Navigating the Maze

In a single-lab project you go straight to the PI.

In Bridge2AI, that straight line becomes a maze. Decisions flow through multiple layers, each with different authorities and responsibilities.



Awareness of this vertical depth prevents wasted time waiting on the wrong person.

Cell Models for AI *Data Generation*

Supported by NIH #1OT2-OD032742-01

cm4ai.org/

People – Data – Ethics



UC San Diego UCSF UNIVERSITY OF SOUTH FLORIDA

UNIVERSITY OF VIRGINIA

Yale

Stanford University

SFU SIMON FRASER UNIVERSITY

TEXAS The University of Texas at Austin

THE HASTINGS CENTER

UAB

CELL MODELS

Anti-cancer therapies

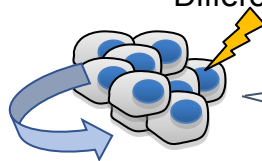


Triple negative breast cancer (TNBC) cells

Paclitaxel

Vorinostat

Differentiation



Human induced Pluripotent Stem Cells (iPSCs)

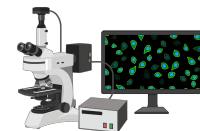


Neural cells

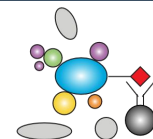


Cardiac cells

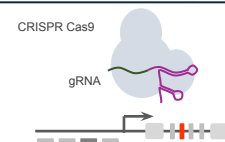
EXPERIMENTAL PLATFORMS



Confocal microscopy



Proteomic mass spectrometry



Single cell RNA sequencing

Why Governance Matters

Multi-disciplinary & Multi-institutional

Large projects bring together specialists from many fields and institutions, increasing coordination challenges and potential conflicts. **Basic science / engineering / bioethics**

Balancing Interests

Governance frameworks aim to balance the interests of diverse partners and ensure the legitimacy of decisions. **Speed to execution vs. collecting broad perspectives.**

People, Policy & Process

Consortia governance spans macro, meso and micro levels: regulators and funders, institutional policies, and local norms and routines. E.g., **Data Governance**

Embedded Contracts

Many macro structures are set in contracts by funders, leaving limited room for negotiation at the project level. **From five module to three pillar transition.**

Clear Rules Prevent Conflict

Up-front agreements on data access, publication, and decision pathways defuse conflicts and promote fairness. **PI vs. committee**

Dynamic & Adaptive

Effective governance is flexible, evolving with changing needs and continuing to embed public good principles. **Rules vs. Situations**

Three Governance Models:

Traditional vs Distributed vs Network

Model	Hierarchical	Distributed	Network
Structure	Clear chain of command	Autonomy & network of leaders	Central hub with spokes
Benefits	Consistency & compliance	Innovation & engagement	Efficient communication & resource alignment
Challenges	Bottlenecks & inflexibility	Requires trust & clear accountability	Hub capacity & limited spoke collaboration

Who holds the Key?





Casey Greene, PhD

Professor and Chair, Department of Biomedical Informatics, the University of Colorado School of Medicine (CU Anschutz)

Working at the interface of deep learning and biology. Expert in modeling complex biological systems. Pioneer in open source biology and deep learning

- B.S. in Chemistry from Berry College.
- Ph.D. in Computational Genetics (2009), Dartmouth
- Postdoctoral, Princeton University (2009–2012)
- PI, Integrative Genomics Lab (2012-15) Dartmouth
- Faculty (2015-2020), Penn School of Medicine
- Founding Director (2020-22), CU Center for Health AI
- Chair (2022-), founding chair of Department of Biomedical Informatics





Fireside Warm-up Questions (Bridge2ai)

Onboarding

What is the bridge2ai governance structures?

Control vs Freedom

In bridge2ai, how do you ensure distributed leadership doesn't blur accountability for data-generation project goals?

Network & Micro-structure

How did the bridge2ai center and DGPs's governance structure evolve over time?

Closing

What governance lessons would you share with new members?



More Leadership Questions with Dr. Casey Greene

- 1.Putting serendipity into practice:** You've said that biomedical informatics is "in the serendipity business," emphasizing the need to put the right information in front of the right person at the right time to support good decisions . How does that philosophy influence the way you structure decision-making processes and governance within a large consortium?
- 2.Connecting diverse stakeholders:** At Colorado you lead a department that bridges basic scientists, clinicians and technologists; collaboration is described as integral to its mission . What governance mechanisms have you found most effective for aligning such diverse stakeholders around shared goals, and how do those approaches differ from your time at Penn?
- 3.Balancing openness and cohesion:** The Greene Lab is known for its open, reproducible and inclusive research culture and for pushing to make software and data freely available . How do you balance that openness with the need for cohesion and clear leadership in a consortium where many parties may have differing policies or incentives?
- 4.Enabling risk-taking in safe environments:** You've noted that a safe, supportive lab environment allows researchers to take the risks necessary for discovery . How do you extend that philosophy to consortium-level governance, where risk-taking must be weighed against accountability and compliance?
- 5.Managing innovation vs. equity:** In discussing AI deployments, you've cautioned that the field risks increasing disparities and have stressed the responsibility to support diverse faculty and communities . How do you integrate equity and inclusion considerations into leadership decisions in consortium governance, and what structures help ensure under-represented voices are heard?
- 6.Lessons from open review initiatives:** The COVID-19 "living review" and other crowd-sourced projects you've led required coordinating dozens of contributors and embracing distributed leadership . What governance lessons did you take from those experiences that apply to big biomedical consortia, especially regarding decision rights and transparency?
- 7.From department chair to PI:** As chair of Colorado's Department of Biomedical Informatics you oversee departmental governance while still serving as a PI. How do you navigate the tension between administrative oversight and scientific creativity, and what advice do you have for early-career PIs facing similar dual roles?
- 8.Scaling trust across layers:** You've highlighted that informatics can connect basic science and clinical practice to "make a real, meaningful difference in the world" . In multi-layered consortia, what strategies build trust up and down the hierarchy—from trainees to funding agencies—so that information flows efficiently and decisions are accepted?
- 9.Future of consortium leadership:** Looking ahead, what emerging challenges (e.g., data privacy, AI regulation, sustainability) do you foresee affecting governance models for large research consortia, and how can current leaders prepare the next generation of scientists to navigate them?